

BUSINESS ANALYTICS, Term I

2 Credits + Revision Hrs

Number of Teaching Hours: 30 + additional 10/12 Hrs of supplementary learning)

Course Description:

Organizations are using analytics strategically to make better decisions and improve customer and shareholder value. The past decade has seen remarkable growth in analytics among business organizations. Core topics in business analytics like statistics, data analysis and information and decision support systems have been a part of business education for long but presented in separate and independent courses with little topical integration. This course is designed to present the emerging discipline of business analytics in a unified fashion consistent with the contemporary definition of the field

Learning Outcomes

Upon completion of this course, students will be able to complete the following key tasks:

1. Gain an understanding of how managers use business analytics to formulate and solve business problems and to support managerial decision making.
2. Gain an understanding of different mathematical and statistical models available for modelling of various business scenarios including decision analysis methods.
3. Use Excel for data preparation and analysis for all units in the course

Level of knowledge

The course essentially deals with data based descriptive and predictive models to aid decision making and will use excel and related add-ins extensively. Therefore, participants in this course are expected to have basic understanding of topics covered in Managerial statistics and should have basic exposure to working with MS excel.

The entire course will be in MS Office Excel with Goal Seek and Solver applications

Unit I Introduction to Business Analytics (4 Hrs)

Introduction to Business Analytics (BA)*. Evolution and Scope of Business Analytics. Data for Business Analytics. Decision Models – Descriptive, Predictive and Prescriptive Models. Problem Solving and Decision-making process.

Unit II Introduction to Data Structuring and BIG DATA in Excel (2 Hrs)

Basic Excel skills. Using Excel functions and developing Spread Sheet Models. Art of developing Spread sheet models

Pivot DATA arrangement and reading

Unit III: Descriptive Analytics: Data Concepts & Data Visualization using Excel 12 Hrs)

Data Concepts: Data, Measurement & scaling

Data Visualization (application-based visualization with theoretical relevance)

(a) Analytical Visualization Tools

- OLAP cubes

(c) Time-Based Visualization

- Gantt chart

(d) Distribution & Pattern Analysis

- Pareto graph / Index estimation
- Histograms
- Scatter / correlation etc

(e) Business Analytics Visuals

- Whale curve
- ABC curve
- Water fall analysis
- LIFT analysis

(f) Financial Visualization

- Break-even point (BEP) and graphing
- Demand Supply lines and price and qty estimation
- Cost curves (economics and implications)

Unit IV Predictive Analytics Methods (Using Excel 12 hrs)

- a) Regression Analysis (Single variable / Multiple variables) data structuring
- b) Forecasting Techniques will include:

- Linear regression
- Categorical forecasting (Dummy variable)
- Moving average
- Exponential smoothing methods
- Trend
- ARIMA

- c) Learning curve effects cost prediction

All the above techniques would be discussed and then done in excel solver

Additional

Case study bases revision on topics covered & revision (10/12 hours)

Recommended Reading

1. Albright C. S., Winston Wayne L. and Zappe C. J (2009). *Decision Making Using Microsoft Excel (India Edition)*. Cengage Learning